

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Pharmacy (B. Pharm)

Semester – VIII University Examination May 2016

PH411 Pharmaceutical Analysis-IV

Date: 16/05/2016, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book is to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q.1 Attempt any TWO of the following:

A Answer the followings:

[1] Give the principle of Mass Spectroscopy. Draw well labeled diagram of double beam Mass spectrometer and explain function of each component. 07

[2] Explain Metastable Ion in detailed 03

B What is Chemical Shift? Write brief note on factors influencing chemical shift. 10

C Write a Short Note on:

[1] Die's Alder and Mc Lafferty Rearrangements along with it's mechanism 05

[2] Spin-Spin Coupling 05

Q.2 Attempt any TWO of the following:

A Answer the followings:

[1] Detector Used in NMR Spectroscopy 05

[2] Shielding and Desheilding effect 05

B Write a Short Note on:

[1] Isotopic Abundance 05

[2] Coupling Constant (J) 05

C Answer the followings:

[1] Explain the Principle of NMR Spectroscopy. 05

[2] Enumerate Applications of MASS Spectroscopy. 05

SECTION- II

- Q 3 Attempt any TWO of the following:**
- A Answer the followings:**
- [1] Draw a neat labelled diagram of X-ray tube. 05
 - [2] Give applications of atomic absorption spectrometry. 05
- B Answer the followings:**
- [1] What is Bragg's law? Derive the equation for Bragg's law. 05
 - [2] Describe in brief about role of graphite furnace in atomic absorption spectrometry. 05
- C Answer the followings:**
- [1] Which are the different analytical techniques that uses X-ray? Explain in brief. 05
 - [2] Write a note on Laue method in XRD. 05
- Q 4 Attempt any TWO of the following:**
- A** Write a note on radiation sources used in atomic absorption spectrometry. 10
- B** Describe in detail about solvent delivery system and detectors used in supercritical fluid chromatography. 10
- C** Explain in brief about Geiger-Muller and Scintillation counter. 10
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